Ex:-8
Date:

To discover live Hosts using Nmap Scans (ACP, ICMP, TCP/UDP) on TryHackMe Platform

Room link

Aim :-

To discover live hosts using Nmap scans (ARP, ICMP, TCP/UDP) on the TryHackMe platform

Algorithm :-

1. Use Nmap to discover systems that are active in a network.
2. Specify targets to be scanned. Targets can be given as list of IP address, range, subnet
3. Nmap can discover live hosts using diff. methods
ARP scan - uses ARP requests to discover live hosts
ICMP scan - uses ICMP requests to identify live hosts
TCP/UDP ping scan - sends packets to TCP & UDP ports to determine live hosts

→ ARP host discovery :-

Used to obtain MAC address of computer with specific IP address

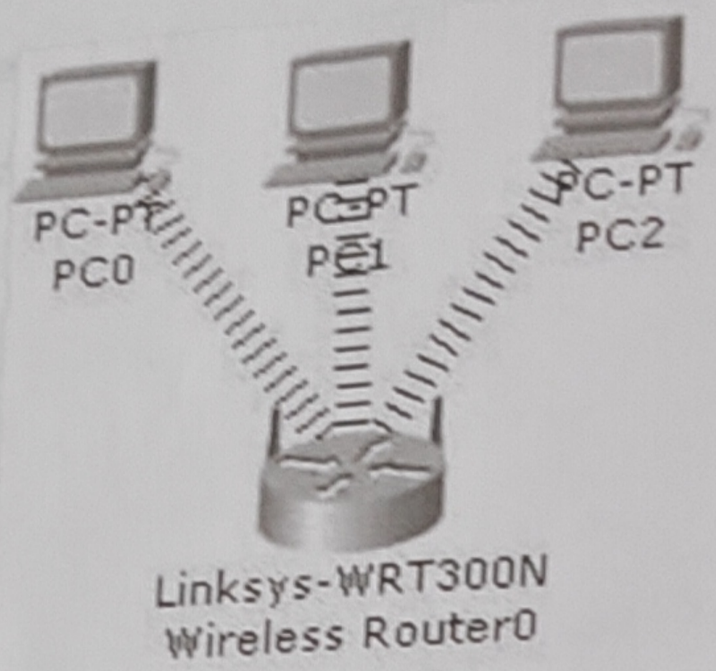
It discover devices within same subnet.

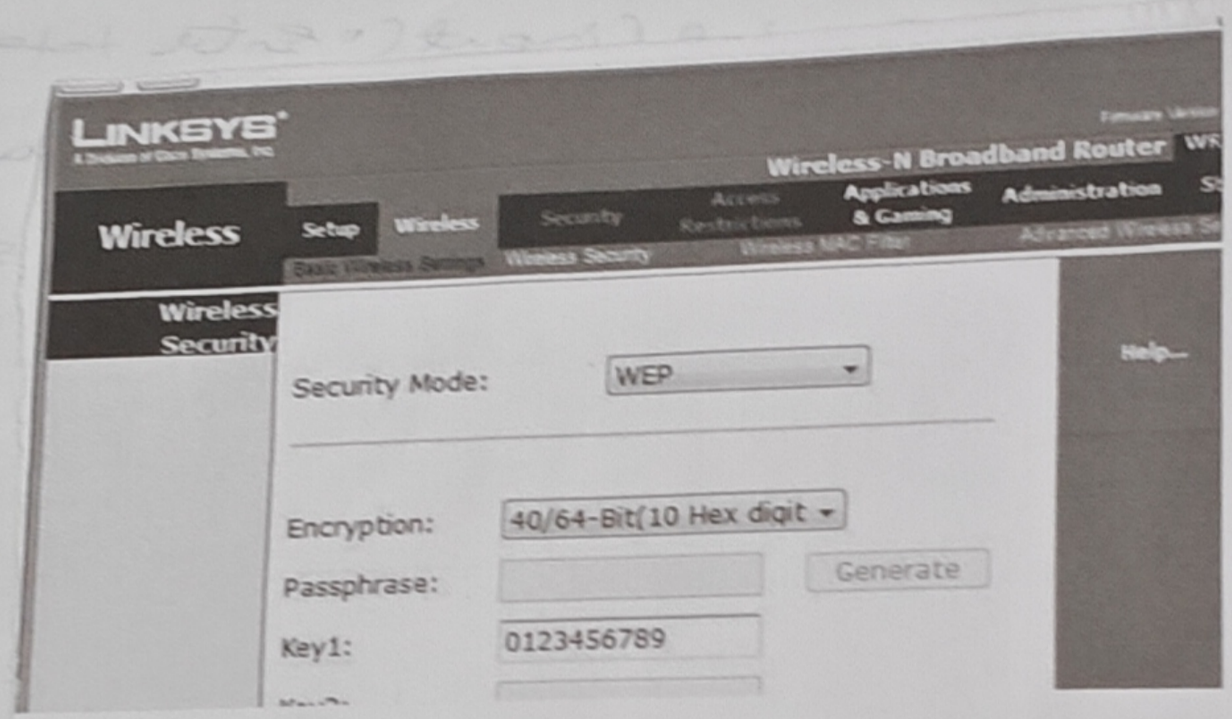
To perform only ARP scan without port scan

nmap -PR -sn TARGETS

PR → ARP scan

-sn → performs host discovery without conducting port scan


PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
Linksys-WRT300N
Wireless Router0


LINKSYS
Wireless-N Broadband Router
Wireless
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Administration
Basic Wireless Settings
Wireless Security
Wireless MAC Filter
Wireless Security
Security Mode:
WEP
Encryption:
40/64-Bit(10 Hex digit
Passphrase:
Generate
Key1:
0123456789
Help...

→ ICMP host discovery :

Can be used to identify live hosts by sending ICMP Echo request & checking for an ICMP echo reply

-PE

other ICMP options

-PP ICMP Timestamp discovery

-PM ICMP address mask discovery

-PE ICMP Echo discovery

→ TCP host discovery :-

TCP SYN Ping used to determine whether a host is active

An open port replies with SYN/ACK

A closed port results in RST

→ UDP host discovery :-

Nmap send packets to UDP ports to determine whether hosts are live. TCP & UDP ping scans used for checking target responses.

Quiz

Result :-

Thus the live hosts successfully discovered using Nmap scans based on ARP, ICMP, TCP and UDP host discovery techniques.